

# Creating image pull secret

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Create an image pull secret to enable the OpenShift® cluster to authenticate with the IBM Entitled Registry. The image pull secret provides the credentials to access the IBM Entitled Registry.

## Before you begin [🔗](#)

- As a prerequisite to run base64, you must install base64 or jq.  
Download the jq JSON processor command line package. To download the jq JSON processor package, see [Download the jq JSON processor command line package](#).  
Combine the JSON value of the default global pull secret with the private registry pull secret that you want to add.
- Ensure that you have access to [https://cloud.ibm.com/docs/openshift?topic=openshift-access\\_cluster](https://cloud.ibm.com/docs/openshift?topic=openshift-access_cluster).

## Procedure [🔗](#)

1. Run the following command to log in to OpenShift system:

```
oc login --token=<API token> --server=https://<server name>:6443
```

2. Create a base64 encoded string of the credentials used to access the image registry.

**Linux example:**

```
echo -n "cp:GENERATED_ENTITLEMENT_KEY" | base64 -w0
```

**Windows example:**

```
echo -n "cp:GENERATED_ENTITLEMENT_KEY" | base64 -e
```

**Note:** Replace GENERATED\_ENTITLEMENT\_KEY with the entitlement key that you generated in [Obtaining entitlement key](#).

3. Create an authority.json to include the base64 encoded string of your credentials (created in the previous step), user name (to access cp.icr.io), and entitlement key for the IBM Cloud Container Registry.

**Windows or Linux example:**

```
{  
  "auth": "BASE64_ENCODED_ENTITLEMENT_KEY"  
}
```

**Note:** Replace BASE64\_ENCODED\_ENTITLEMENT\_KEY with the value of base64 encoded entitlement key got from the previous step.

4. The following step takes the authority.json and includes it as a new authority in your .dockerconfigjson, stored as a temp\_config.json.

```
oc get secret/pull-secret -n openshift-config -ojson | jq -r '.data[".dockerconfigjson"]' | base64 -d | jq '.[].cp.icr.io' += input' - a
```

**Note:** The referenced jq command is not installed on OpenShift Container Platform clusters by default. Run the following command to install jq.

```
yum install jq
```

The yum install jq command is for Red Hat® Enterprise Linux systems. For other operating system, use the appropriate commands to install jq.

5. Use the contents of the temp\_config.json file and apply the updated config to the OpenShift cluster.

```
oc set data secret/pull-secret -n openshift-config --from-file=.dockerconfigjson=temp_config.json
```

6. To verify whether your pull-secret is updated with your new authority, run the following command and confirm that your authority is present.

```
oc get secret/pull-secret -n openshift-config -ojson | jq -r '.data[".dockerconfigjson"]' | base64 -d
```

The updated config is now rolled out to all the nodes in the OpenShift cluster.

7. When the global pull secret is updated, enter the following command to remove the temporary files that were created.

```
rm authority.json temp_config.json
```

**Parent topic:**

→ [Prerequisites](#)